

CSCI-UA-060

Database Design & Web Implementation



Course Topics

Core Database Principles

RDBMS (Relational Database Management System)
SQL (structured query language)
NoSQL (not only SQL)

Web Implementation

Front-end/Back-end development
Dynamic Interfaces
HTML Templating

Frameworks

SQLite
MongoDB

Frameworks

Python -> SQLite
Flask - Jinja

Additional Tools

EdStem Workspaces
LucidChart
Github
SQLite DB Viewer
VS Code

Course Schedule Overview:

Database Normalization
Creating SQL Databases
Populating SQL Databases
Querying SQL Databases
Web Implementation -
Python/Flask/SQLite
MongoDB
Web-Python-mongitaDB

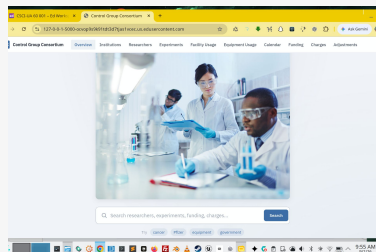
Assessment

Quizzes + Class
Participation : 10%
Homework: 30%
Group Project: 20%
Midterm/Final: 40%

final project

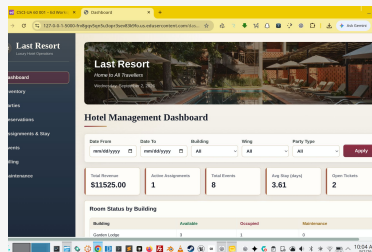
A semester long group project developing a functioning, dynamic data driven web application integrating an SQLite backend.

SOME PAST EXAMPLES



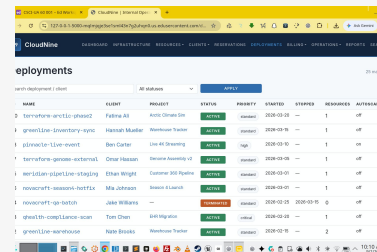
CONTROL GROUP CONSORTIUM

<https://edstem.org/us/courses/91931/workspaces/p2dqMOEkbiw76xoG5sII2ZaQpMx4fBf>



LAST RESORT HOTELS

<https://edstem.org/us/courses/91931/workspaces/pvgmzOZm9c4hQWApOhAjon6h8wxjaSPZ>



CLOUD NINE DATA SERVERS

<https://edstem.org/us/courses/91931/workspaces/paNEVO7EgLHYmOZ0pbpiMyNj2jfPM6dQr>

in-class
workshops

**group workshop
sessions submitted
by the end of class**

Dictionary in Python

PNative.com

Unordered collections of unique values stored in (Key-Value) pairs.

```
d = {'a': 10, 'b': 20, 'c': 30}
```

d['a']
d['b']
d['c']

- ✓ Unordered: The items in dict are stored without any index value
- ✓ Unique: Keys in dictionaries should be Unique
- ✓ Mutable: We can add/Modify/Remove key-value after the creation

List in Python

PNative.com

```
L = [20, 'Jessa', 35.75, [30, 60, 90]]
```

L[0]
L[1]
L[2]
L[3]

- ✓ Ordered: Maintain the order of the data insertion.
- ✓ Changeable: List is mutable and we can modify items.
- ✓ Heterogeneous: List can contain data of different types
- ✓ Contains duplicate: Allows duplicates data

Tuples in Python

PNative.com

```
T = (20, 'Jessa', 35.75, [30, 60, 90])
```

T[0]
T[1]
T[2]
T[3]

- ✓ Ordered: Maintain the order of the data insertion.
- ✓ Unchangeable: Tuples are immutable and we cannot modify items.
- ✓ Heterogeneous: Tuples can contain data of different types
- ✓ Contains duplicate: Allows duplicates data

Arrays like lists, dictionaries, tuples, live in-memory.
Memory disappears when the program stops running

Starter data

```
students = [  
    [1, "Alice", "GSAS", 32],  
    [2, "Bob", "Tandon", 28],  
    [3, "Carol", "GSAS", 40],  
    [4, "Dan", "CAS", 18]  
]  
# columns: [id, name, school, credits]
```

Tasks

- 1 Print all student names (one per line).
- 2 Print only the students in GSAS (id and name).
- 3 Print the students with credits > 30 (name and credits).
- 4 Insert a new student row for: id=5, name='Eve', school='CAS', credits=22.
- 5 Update Bob's credits to 30.
- 6 Reflection (comment): What breaks if someone inserts a new column in the middle?

homework assignments

individual

weekly and

bi-monthly

projects

SOME PAST EXAMPLES

The screenshot shows a GitHub repository page for 'Assignment 1 - FileBasedDataSystems'. The main content is a PDF document titled 'Asking Questions with CSV Data'. The document is 79.5 KB and is described as 'Conceptual Data Exploration with Native Python or Pandas'. The 'Goal' section states that the assignment emphasizes conceptual data literacy over advanced Python syntax. The 'Dataset' section mentions choosing a CSV dataset from NYC Open Data or Kaggle, with a link to <https://opendata.cityofnewyork.us/data/>.

The screenshot shows a GitHub repository page for 'Assignment 3 - Queries'. The main content is a PDF document titled 'Assignment 3 Requirements.pdf'. The document is 99.1 KB and contains 'Overall Requirements' and 'Specific Requirements'. The 'Overall Requirements' section states that students will develop SQL queries to respond to questions listed below concerning the United Helpers activities. The 'Specific Requirements' section lists three tasks: 1) Importing the database, 2) Answering questions, and 3) Running a script.

The screenshot shows a GitHub repository page for 'Assignment 7 - Flask + SQLite Bookstore'. The main content is a PDF document titled 'Assignment 7 - Flask + SQLite Bookstore.pdf'. The document is 339 KB and contains an 'Overview' section. The 'Overview' section states that the assignment is a handwritten Parts Order Form for a local supply company. The 'Specific Requirements' section lists five tasks: 1) Connecting Python to a database, 2) Executing SQL queries from Flask, 3) Rendering dynamic pages using templates, 4) Organizing a back-end and front-end within the same project, and 5) Using SQLite inside EdStem.

The screenshot shows a GitHub repository page for 'Assignment 4 - Normalization'. The main content is a PDF document titled 'Assignment4.pdf'. The document is 377 KB and contains 'Specific Requirements'. The 'Specific Requirements' section lists five tasks: 1) Connecting Python to a database, 2) Executing SQL queries from Flask, 3) Rendering dynamic pages using templates, 4) Organizing a back-end and front-end within the same project, and 5) Using SQLite inside EdStem.

Part Number	Name	Type	Case Code	Quantity Ordered	Unit Price
1043M	Front Control	Plumbing	6413	4	12
1043M	Module for	Electrical	H 433	3	7
10776	Flow Assembly	Plumbing	6413	7	9
10657	Flow Assembly	Plumbing	6413	5	10

reading quizzes

**taken on
brightspace,
submitted anytime
before deadline**

Part of a reading series

Taken online via Brightspace and Respondus asynchrono usly

Edit Quiz - Database Design x +

brightspace.nyu.edu/d2l/le/activities/edit/TmpZd05sODFNVEF3TUY4ME9USTB...

Edit Question Pool

Multiple Choice

The sequence of Normalization levels is:

Multiple Choice

In most cases, the goal of normalization is to reach:

Multiple Choice

A partial dependency occurs when:

Multiple Choice

A Transitive Dependency occurs when:

Save Cancel

Randomly selecting 10 questions from 11

A Transitive Dependency is occurs when:

- ☐ There is no primary key, so attributes are not dependent on each other.
- ☐ A non-key attribute is only dependent on part of the primary key
- ☐ One part of the primary key is dependent on another part of the primary key
- ☐ A non-key attribute is

6:05 PM 9/2/26

exams

**midterm (database
normalization, sql +
python basics)**

**final exam (joins,
complex sql queries,
python/flask,
mongoDB)**

Midterm Overview - Google Slides

File Edit View Insert Format Slide Arrange Tools Extensions Help

Search + Add Undo Redo Find Fit Background Layout Theme Transition

1 Midterm Overview

2 Midterm Logistics

3 Midterm Exam Questions

4 Section 1

5 Google Python Lists/Dictionaries

6 Midterm Review

Midterm Exam Structure

There will be no multiple choice questions. All questions will require short responses.

Two sections:

Section 1	Section 2
10 Questions: Python Lists and Dictionaries Normalization PK's and FK's Relationships DDL and DML Commands practiced in class	5 Questions: Modelling Tables and Attributes PK's and FK's Relationships Datatypes Select Statements

6:13 PM 9/2/26

Final Prep - Google Slides

File Edit View Insert Format Slide Arrange Tools Extensions Help

Search + Add Undo Redo Find Fit Background Layout Theme Transition

1 Database Design and Implementation

2 Overview

3 Topics Included

4 SQL Questions

5 SQL Questions

Topics Included

- SQL DML statements (INSERT, UPDATE and DELETE)
- SQL SELECT statements, including
 - JOINS (inner/left)
 - GROUP BY (with functions)
 - using SUB-QUERIES
- Using Python to access:
 - SQLite
 - MongoDB - basic statements only
- Using Flask/JINJA2
- HTTP GET/POST

Click to add speaker notes

6:24 PM 9/2/26

EdStem



EDSTEM

DB Browser SQLite



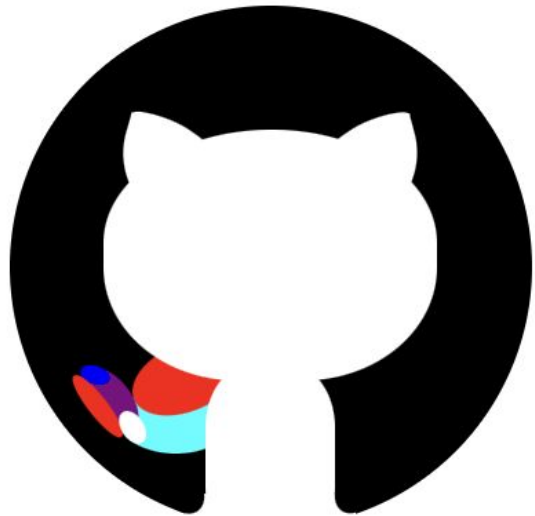
SQLite Browser

A screenshot of the SQLite Browser web application. The interface shows a table with columns 'AlbumId', 'Title', and 'ArtistId'. The table contains 8 rows of data. There are search boxes for each column and an 'Upload another file' button.

AlbumId	Title	ArtistId
1	For Those About To Rock We Salute You	1
2	Balls to the Wall	2
3	Restless and Wild	2
4	Let There Be Rock	1
5	Big Ones	3
6	Jagged Little Pill	4
7	Facelift	5
8	Warner 25 Anos	6

Open SQLite Database in Browser

<https://sqlitebrowser.org/>



GitHub

<https://github.com/>
